Image Forgery Detection Using Deep Learning Framework

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***Abstract*—With the continuous development of digital media and the rapid change of digital image processing equipment, digital images are easy to be tampered with by criminals. In recent years, illegal elements often use tampering images to commit crimes, so it is urgent to study image forensics technology. Blind forensics technology is used to verify the original and authenticity of the image, and then the tampering mode of the image is discriminated and the tampered area is located according to the tampering trace of the image. This paper focuses on the detection of image manipulation and tampering and the localization of tampered areas.The main innovation of this paper is to use the core idea of end-to-end training of u-net network to further optimize the performance of image tampering detection tasks in combination with residual network.**

Keywords—digital images, dynamic convolution, image classification, feature representation, residual network

I. Introduction

With the rapid popularization of the Internet and the rapid development of modern information technology, digital images and videos have become the main media of people's social and entertainment. In view of the unreliability of human eyes in identifying false images, automatic and accurate forgery detection schemes are crucial in many applications. Researchers in academia and industry have designed three general image authentication methods: digital signature, image watermark and forensics analysis. Image restoration is a common image processing method that uses image processing technology to repair damaged or degraded areas in an image. [1]It is often used to repair old photos and remove watermark, date and other information in pictures. Image restoration restores the true vision of the image on the basis of maintaining the harmony and unity of the original image. In daily life, people may use image restoration algorithms to erase some places in the image for entertainment, resulting in false images.[2] Image restoration itself is not designed for image tampering, but it can not avoid some people with ulterior motives applying existing image restoration algorithms to image tampering, such as removing the original objects or people in the image in an attempt to change the objective facts reflected in the original image. Image stitching is one of the most common types of image forgery. It copies a specific area of one image (called donor image) and then pastes the area onto another image (called host image). Before pasting, a specific area can be geometrically transformed by rotation and scaling to match the host image.

Image stitching may be maliciously applied in real life, such as forging evidence in court, forging bank electronic statements, reporting false news to the public, etc., which may lead to serious security problems. Copy move forgery is a common method of image tampering, which is to copy a certain area of an image, and then paste it into other areas of the same image after scaling, rotation and other post-processing operations, so as to hide specific objects in the image or add objects that do not exist in the image. If copy move images are used to report false news, forge evidence or paper data, it will have a negative impact on social stability, judicial justice and academic atmosphere. Therefore, it is necessary to authenticate the authenticity of the copy move image and locate the tampered area in the image.The following figure 1 shows the photos taken when the wild South China tiger appeared. When the photos were released, people had a heated discussion. Some people were very happy about the "sudden appearance" of the wild South China tiger and asked the country to protect it. Others think that the picture is forged. After technical screening by experts, they found that the picture was forged.

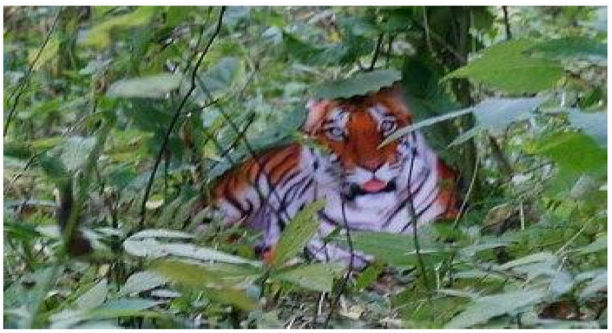

Fig.1 Forged picture of South China tiger

II. Related works

A. Repair forged image

Repair forgery refers to the process of reconstructing the missing or damaged parts of an image.[3,4,5] Image restoration can be used to repair old movies, restore degraded old photos, and remove watermarks, dates and other information in pictures. Image restoration restores the true vision of the image on the basis of maintaining the harmony and unity of the original image. Image restoration itself is not designed for image tampering, but it can not prevent some people with ulterior motives from maliciously tampering with the image by using image restoration algorithms, such as removing the original people or objects in the image in an attempt to change the objective facts reflected in the original image.

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B. Image tampering Technology

Image tampering mainly aims at modifying image content to achieve malicious purposes such as deception and forgery Both manual tampering technology and automatic tampering technology follow the characteristics of the basic types of image tampering. Image tampering is mainly divided into synthesis, embellishment, computer generation, variation, enhancement, painting and so on. [6]Synthesis is to synthesize partial areas of different images into the same image, resulting in erroneous visual effects. The synthesizing operation also includes moving part of the image to other areas of the image to hide the objects in the area or cause content ambiguity, Common retouching operations include face grinding and contour stretching. Computer generation refers to the generation of images with a specified style through feature parameters that can be recognized by computers. [7]The scenes and objects in the images do not exist in reality, but visually conform to the laws of reality. Variants are generated by targeting The learning of image features gradually changes other images into other images through iteration. [8]The variation process is actually a process in which the distance between the feature vectors of some aspects of the two images is constantlynarrowing, and the two images are gradually fused into a new image through different weights. Enhance and change parameters such as light, brightness, hue and contrast of the image, highlight some areas of the image or reduce the visual attention of some areas. Enhancement operations are often used to mask the specified content of an image or erase tampering traces. [9,10,11,12]Painting uses a brush tool to paint on an image. The painting content determines the tampering quality according to the level of the operator. Painting tampering can achieve the tampering goal of forging or covering a specific object.

C. Image tampering detection method

The process of image tampering detection is also the process of tampering forensics. Forensics technology can be divided into forensics technology based on manual design features and forensics technology based on deep learning. Forensics technology based on manually designed features usually needs to design features for some characteristics of tampered images, calculate the image tampering score through features, and identify tampering according to whether the score reaches the threshold; The forensics technology based on deep learning can identify the deep features in images by designing a reasonable deep learning network, improving the learning ability of the network, guiding the learning direction of the network, and adaptively detect the historical tampering behaviors contained in images through the learning and training of a large number of samples.Traditional image tampering detection methods are often not independent of each other and can be used interchangeably. [13,14,15]However, these methods usually extract features by manual design, so most of them have limitations and lack of representativeness. They can not judge multiple tampering methods at the same time according to these features, resulting in traditional image passive forensics methods can only identify a specific tampering type. With the continuous development of image tampering means, traditional algorithms can no longer be applied to the detection of a large number of forged images with complex processes from the perspective of generalization and timeliness, and tampering features are increasingly hidden and difficult to locate. [16]Therefore, most of the current research on forensics focuses on deep learning. Deep learning methods can regard image passive forensics as a problem of target detection, anomaly detection and even semantic segmentation. [17]In recent years, more scholars hope to make use of the adaptability of deep learning methods to make deep learning models.

Effective features can be automatically extracted. In order to avoid extracting image semantic features, researchers are inspired by traditional image forensics methods and adopt the method of extracting features at the image block level. Before extracting features, first divide the image into image blocks of the same size, and then extract the tampering features of the image blocks one by one (such as manual design features, image statistical features, etc.), avoiding that there are complete semantic subjects in a single image block, At the same time, the processing of image blocks also improves the efficiency of the model.

D. Convolutional neural network

Convolutional neural networks commonly used at present are usually composed of continuous convolution layer, batch normalization layer, nonlinear activation function, pooling layer and full connection layer. As one of the representative algorithms of deep learning, convolutional neural networks are used to process grid structure data, such as time series data and image data.In the field of computer vision, feature extraction and classification have always been an inseparable part. In traditional image processing, the methods of feature extraction are usually designed in advance based on statistical laws or some prior knowledge. This manually designed feature extraction method can not completely extract the information of the original image, and its robustness is not strong. Convolutional neural network is a neural network specially used to process image data, which has a similar network structure. Convolutional neural network adds the concepts of local receptive field, sparse weight and parameter sharing. These three concepts make convolutional neural networks have certain translation and scale invariance compared with other neural networks, and are more suitable for image data learning. Translation and scale invariance ensure that the recognition of specific features is more effective.

III. METHODS

A. Residual network

Convolution networks can only solve linear problems. After scholars add nonlinear activation functions to them, the nonlinear expression ability of the networks is enhanced. In addition, the feature extraction ability of the models can be improved by increasing the depth of the networks. However, he Kaiming of Microsoft Asia Research Institute and others found that the detection accuracy of the model does not improve with the increase of the number of layers of the network. When the number of layers of the network increases to a certain extent, the model degrades. The reason is that with the increase of the number of layers of the network, the loss of the training set will gradually decrease and then tend to saturation. After increasing the network depth again, the loss value will increase.

Multiple shallow networks are superimposed to form a deep network: ResNet is constructed by superimposing multiple mappings on the composition of the shallow

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network, so the performance of ResNet with deep layers may not be lower than that of the shallow network; However, like convolution in traditional CNN, ResNet is difficult to construct such an identity mapping block by directly increasing the number of layers; After the ResNet shallow network reaches saturation, the identity mapping layer is added immediately, so the number of layers of the network will deepen, and the performance of the network will be enhanced, so as to ensure that the error will not increase with the depth of the ResNet model.

The addition of identity mapping blocks changes the learning objectives. In the training process, there is no need to learn the final output data, but only the difference between the input data and the output data, which is the residual. The data corresponding to the two input-output paths are combined and combined into a residual block through the action of a nonlinear activation function. The side branch of RESNET is to directly connect the output of the previous layer to the input of the next layer or to more layers. This connection.This method is called shortcut or jump connection.

However, the depth of the network is not as deep as possible In a certain depth range, with the increase of network layers, the model can fit more complex functions, and the performance of the model can be improved However, after the number of network layers increases to a certain number, the training accuracy and test accuracy decrease rapidly Researchers have found that with the increase of network depth, the effect of increasing the depth of the network after the accuracy reaches saturation becomes worse and worse.

The deep residual network is composed of multiple residual learning units given the input image data, the depth residual network first sends the input data to the convolution layer conv, the nonlinear activation function layer relu and the batch normalization layer batchnorm in sequence; Then the processed results are further sent to multiple residual units, and then subjected to batch processing normalization layer BN and multiple full connection layer; Finally, the output result is obtained.On the basis of the original residual element, the placement and combination of the batch normalization and activation function relu, as well as the different construction methods of the residual element, have an impact on the accuracy, the convergence of the network and the speed of training

B. U-net

U-net is composed of the coding block on the left side and the decoding block on the right side, in a symmetrical structure. The coding decoding layers are hop connected. The coding block is composed of convolution.The convolution layer extracts image features and obtains a preliminary effective feature map. The pooling layer increases the receptive field and improves the global information acquisition ability by reducing the resolution of the feature map. The coding block is composed of five modules, each of the first four modules is composed of one convolution layer and one pooling layer, and the fifth module is composed of three convolution layers and three upsampling pooling layers. The decoding block is composed of an upscale pooling layer and a convolution layer, in which the upsampling pooling layer is used to restore the resolution of the feature map. The decoding block is composed of four modules, each of the first three modules is composed of four convolutional layers and two upsampling pooling layers, and the fourth module is composed of two convolutional layers. The hop connection between the encoding and decoding layers stitches the feature map extracted from the convolution layer of the encoding block and the feature map output from the upscale pooling layer in the decoding block, and integrates the position information of the bottom layer and the image information of the top layer.

The core idea of u-net is to use less training sets for end-to-end training. The u-net model is based on the fully connected convolutional neural network (FCN) architecture, which modifies and designs the FCN structure more finely. However, the network can not recover the detailed structure information of the image after the pooling layer, which is easy to cause segmentation fracture. Based on this, this study will add the edge zero filling and BN (batch normalization) layer on the basis of the U-net model. By performing the edge zero filling operation on the image in the convolution process, the network will better consider the edge pixels of the image; At the same time, the size of the image remains the original size after convolution, and the image resolution is changed only when the image features are pooled with the maximum pooling layer. Adding the BN layer is mainly to normalize the mean and variance of the output data of each channel, so that the mean and variance are close to 0 and 1, so as to improve the key indicator F1 and better improve the network segmentation performance.This paper proposes to replace the feature extraction module with the residual network module. From the follow-up experiments, it can be seen that the method proposed in this paper has played a certain role in promoting the effect of image tampering detection task.

IV. EXPERIMENTS

A. Experimental setup

The relevant experiments of this study were implemented on the Ubuntu 16.4 system, geforce GTX 1080ti GPU, adopting the pytoch framework and adaptive moment estimation (Adam) as the optimizer. The learning rate was initialized to 1 x 10-3, set to 1 x 10-4 after 30 rounds of training, and set the batch size to 8. A total of 300 rounds of training were conducted. Finally, the model with the highest F1 score and precision for the test set was selected as the test model.

B. Ablation study

The methods in this paper are all based on deep learning. The purpose is to test whether the proposed method can accurately detect whether the input image is a tampered image, classify the tampered image and locate the tampered area. Therefore, when selecting the evaluation indicators, this paper selects the precision and F1 score as the evaluation indicators of this study. Precision refers to the prediction result, which means the probability of actually being a positive sample among all the predicted positive samples.F1 score is equivalent to the combination of recall and precision, and can also be regarded as the weighted average of recall and precision.The data set used in the model in this paper is casiav2.0. In order to ensure the practicability of the model, the data set contains not only people, animals and plants, but also various natural scenery and national buildings. In addition, Columbia dataset are also used, which are close to nature and life. Columbia is a data set captured and synthesized with a camera.From the experimental results in

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table I, the F1 score and average accuracy of the proposed method in casiav2.0 are 0.7116 and 0.8891, respectively, and the F1 score and average accuracy in the Columbia dataset are 0.9028 and 0.9533, respectively. The F1 score and average accuracy of the classical CNN based method(DCT) in casiav2.0 are 0.498 and 0.349[18], respectively, and the F1 score and average accuracy in the Columbia dataset are 0.463 and 0.365, respectively. From the experimental results, it can be seen that the end-to-end image intrinsic attribute segmentation network can achieve forgery detection without any preprocessing and post-processing. Compared with the traditional detection method based on feature extraction and the existing detection method based on CNN, it can obtain better and stable detection performance. The method proposed in this paper has played a certain role in promoting the effect of image tampering detection task.

TABLE I. COMPARISON RESULTS ON COLUMBIA DATASET AND CASIAV2.0

Model	**Columbia dataset**		**casiav2.0**	
	F1 score	***average accuracy***	***F1 score***	***average accuracy***
CNN[17]	0.463	0.365	0.498	0.349
Ours	**0.9028**	**0.9533**	**0.7116**	**0.8891**

V. CONCLUSION

This paper introduces the basic concept, research background, technology introduction, the environment used in the experiment, the data set used in the experiment and so on. And the main work of this paper is introduced in detail: the fusion of u-net and residual network is carried out to analyze the experimental performance of the new method. From the experimental results, it can be seen that the method proposed in this paper has played a certain role in promoting the effect of image tampering detection task.

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